

Product Requirements Document

ChatAssist — TechViet Internal AI Knowledge Assistant
Product Management Team

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1. Executive Overview

ChatAssist is an AI-powered internal knowledge assistant that enables TechViet employees to instantly find information from internal documentation, policies, technical wikis, and knowledge bases using natural language queries. The system uses Retrieval Augmented Generation (RAG) to provide accurate, sourced answers grounded in TechViet's actual documentation — eliminating hallucination risks associated with pure LLM approaches.

This project was initiated by the CTO based on findings from the 2025 Employee Experience Survey, where "difficulty finding internal information" was the #2 pain point (after "meeting overload"). The project has executive sponsorship from the CEO and a dedicated cross-functional team of 6 (2 ML engineers, 2 backend engineers, 1 frontend engineer, 1 product manager).

1.1 Problem Statement

TechViet employees currently spend an estimated 5.2 hours per week searching for internal information across 8+ systems (Confluence, Google Drive, Slack history, HR Portal, GitHub wikis, email archives, shared folders, and LMS). This results in approximately 2,380 lost productivity hours per week company-wide (458 employees × 5.2 hours). At an average loaded cost of USD 22/hour, this represents approximately USD 52,000 per week or USD 2.7 million per year in lost productivity.

New employees are disproportionately affected: our onboarding survey shows they spend 8.3 hours/week searching for information in their first 3 months, and cite "not knowing where to find things" as the primary frustration. The average time-to-productivity for new hires is 90 days, and we believe ChatAssist can reduce this to 60 days.

1.2 Success Metrics

Metric	Current Baseline	3-Month Target	6-Month Target	12-Month Target	Measurement Method
Time to find info	15 min avg	< 3 min	< 2 min	< 1 min	In-app analytics (query to answer)
Employee satisfaction (info access)	3.1/5.0 (2025 survey)	3.8/5.0	4.2/5.0	4.5/5.0	Quarterly pulse survey
IT/HR FAQ ticket volume	120 tickets/week	90/week (-25%)	60/week (-50%)	30/week (-75%)	Jira Service Desk analytics
New hire time-to-productivity	90 days	75 days	60 days	45 days	Manager assessment at milestones
Answer accuracy (human-rated)	N/A	> 82%	> 88%	> 93%	Weekly sampling: 50 queries human-rated
Daily active users	N/A	100+ (22%)	200+ (45%)	350+ (78%)	Application analytics (unique users)
Query volume	N/A	500+/day	1,000+/day	2,000+/day	Application analytics
User satisfaction per query	N/A	> 3.5/5	> 4.0/5	> 4.3/5	In-app thumbs up/down + periodic survey

2. Functional Requirements

ID	Requirement	Description	Pri orit y	Spri nt	Acceptance Criteria
FR-001	Chat Interface	Web-based conversational UI with message history, typing indicators, and source citation display	P0	1-2	User can type question, receive streaming response with clickable source links
FR-002	RAG Pipeline	Retrieval-augmented generation with semantic search, reranking, and grounded answer generation	P0	2-4	Answers include relevant source passages; hallucination rate <5% on test set
FR-003	Document Ingestion	Support PDF, DOCX, PPTX, Markdown, HTML, and plain text. Handle tables, images with OCR, and headers.	P0	1-3	All supported formats parsed correctly; table data preserved; >95% text extraction accuracy
FR-004	Source Attribution	Every answer must cite specific documents with page/section references. Clickable links to originals.	P0	3-4	Each answer has 1-5 source citations; clicking opens original document at relevant section
FR-005	Multi-turn Conversation	System maintains context across messages within a session. Handles follow-up questions and pronoun resolution.	P1	4-5	System correctly resolves 'it', 'that policy', 'the same team' in follow-up questions
FR-006	Feedback System	Thumbs up/down on each response. Optional free-text feedback. Feedback stored for model improvement.	P1	5	Feedback captured and visible in admin dashboard; weekly feedback report auto-generated
FR-007	Admin Dashboard	Document management (upload, delete, re-index), usage analytics, query analytics, feedback review	P1	5-6	Admin can upload docs, see top queries, review low-rated responses, trigger re-indexing
FR-008	Access Control	Role-based document access. Users only see answers from documents they're authorized to access.	P0	3-4	HR-only docs not surfaced to non-HR users; access synced with Google Workspace groups
FR-009	Slack Integration	Query ChatAssist via Slack DM or /chatassist command in any channel. Thread-based responses.	P2	7-8	Users can query from Slack; responses appear in thread with sources; supports follow-ups
FR-010	Auto Re-indexing	Daily scheduled re-indexing of connected document sources. Incremental updates for changed/new docs.	P1	4	New docs indexed within 24 hours; changed docs updated; deleted docs removed from index
FR-011	Multilingual	Support Vietnamese and English queries and documents. Auto-detect query language.	P1	6-7	Equivalent accuracy for VN and EN queries; mixed-language docs handled correctly
FR-012	Conversation Export	Export conversation as PDF or shareable link (read-only, with expiry).	P2	8	Exported PDF matches conversation; shared links expire after 7 days

3. Non-Functional Requirements

ID	Category	Requirement	Target	Measurement
NFR-001	Performance	Time to first token (streaming response)	< 1.5 seconds	P95 measured by Datadog
NFR-002	Performance	Full response generation time	< 6 seconds for typical queries	P95 measured by Datadog
NFR-003	Performance	Document ingestion throughput	100 pages/minute	Benchmark with standard test corpus
NFR-004	Availability	System uptime	99.5% (planned maintenance excluded)	Datadog SLO monitoring
NFR-005	Scalability	Concurrent active users	100+ simultaneous sessions	Load test with k6
NFR-006	Scalability	Document corpus size	50,000+ documents (10M+ chunks)	Performance test at target scale
NFR-007	Security	Data residency	All data in Vietnam (AWS ap-southeast-1)	AWS resource tagging audit
NFR-008	Security	Audit logging	All queries, admin actions, and access logged	Centralized in Datadog; 1-year retention
NFR-009	Compliance	Accessibility	WCAG 2.1 AA compliance	Automated scan (axe) + manual audit
NFR-010	Reliability	Data durability	Zero data loss for indexed documents	Qdrant replication factor 2; daily backup

4. System Architecture

Component	Technology Choice	Deployment	Purpose & Notes
Chat Frontend	React 18 + TypeScript	S3 + CloudFront	Responsive chat UI with streaming support, markdown rendering, source citation cards
API Gateway	Kong 3.6	EKS (2 replicas)	Rate limiting (50 req/min per user), JWT validation, request routing, CORS
Backend API	Python 3.12 + FastAPI	EKS (3 replicas, HPA)	Query processing, RAG orchestration, session management, admin endpoints
Query Understanding	Custom NLP pipeline	Within Backend API	Query classification, intent detection, query rewriting for better retrieval
Embedding Model	multilingual-e5-large (560M params)	EKS (GPU node, 2 replicas)	Generates 1024-dim embeddings for documents and queries. Multilingual support.
Reranker	bge-reranker-v2-m3	EKS (GPU node)	Cross-encoder reranking of top-50 candidates to top-5. Significantly improves relevance.
Vector Database	Qdrant 1.8 (self-hosted)	EKS (3-node cluster, NVMe SSD)	Stores ~10M vector embeddings. HNSW index. Payload filtering for access control.
Document Store	PostgreSQL 16 + S3	RDS (Multi-AZ) + S3	Metadata, user sessions, feedback in Postgres. Raw documents and chunks in S3.

Component	Technology Choice	Deployment	Purpose & Notes
LLM Provider	Claude 3.5 Sonnet (Anthropic API)	External API	Answer generation with retrieved context. 200K token context window. Streaming enabled.
Document Processor	Custom pipeline (Python)	EKS (async workers)	PDF/DOCX/HTML parsing, semantic chunking, embedding generation, Qdrant upsert
Task Queue	Redis 7.2 Streams	ElastiCache (cluster mode)	Async document processing jobs, scheduled re-indexing triggers
Monitoring	Datadog (APM + Logs + Custom Metrics)	SaaS	E2E latency tracking, LLM cost monitoring, retrieval quality metrics, user analytics

5. Project Timeline

Milestone	Date	Key Deliverables	Dependencies	Exit Criteria
M1: Foundation	Dec 15, 2025	Document ingestion pipeline, vector store, basic embedding service, S3 storage	AWS infra provisioned (DevOps)	500+ docs ingested; embedding quality validated
M2: Core RAG	Jan 31, 2026	Query pipeline, RAG orchestration, basic chat UI, streaming responses	M1 complete; Anthropic API access	Demo: 10 test queries answered correctly with sources
M3: Security & Quality	Feb 28, 2026	RBAC, document permissions, reranker, query understanding, feedback system	M2 complete; IT security review passed	Access control verified; reranker improves MRR by >15%
M4: Internal Beta	Mar 31, 2026	Beta launch with 50 users (HR, IT, Engineering leads). Monitoring and alerting.	M3 complete; QA sign-off; 2,400+ docs indexed	50 beta users; >82% accuracy; <3s TTFT; feedback loop active
M5: Integrations	Apr 30, 2026	Slack bot, admin dashboard, analytics, auto-reindexing, conversation export	M4 feedback incorporated; Slack admin approval	Slack bot functional; admin can manage docs; analytics dashboard live
M6: GA Launch	May 31, 2026	Company-wide rollout, multilingual support, performance optimization, documentation	M5 complete; load test passed; executive sign-off	Available to all 458 employees; >88% accuracy; SLA met for 2 weeks